

HOME SERVER

Documentation

27.03.2024r

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1. Physical Device

Server: HP T630

Ram: 8GB DDR4

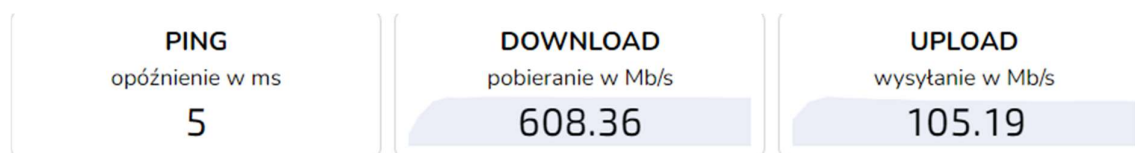
Drive: 256GB SSD

Operating system: Proxmox Virtual Environment 7.4-17

The device can be expanded in the future with more RAM and additional M.2 drive.

Internet: Optical fiber T-Mobile

Speed from <https://www.speedtest.pl/>:



Router: Home Box Wi-Fi 6 Dual-Band (2.4 GHz/ 5 GHz) Multigigabit Ethernet

IPv4 addressing in range 192.168.1.2 – 192.168.1.254

2. Software

The server has the Proxmox Virtual Environment operating system version 7.4-17 installed. It is located at the local address 192.168.1.66. It was installed with the intention of possibly installing additional operating systems in the form of virtual machines. Currently, however, there is only a virtual machine with the Ubuntu Server operating system version 22.04.4.

3. Ubuntu Server 22.04.4

It contains basic applications for managing and sharing websites and web applications. This system was installed to become a hosting for websites and web applications in the future. Currently, several services have been installed on it, such as:

1) Apache2

The service providing websites and web applications in version 2.4.58 also allows the configuration of the HTTPS encrypted module along with adding our certificate. I used the solution of placing the website files in the "public_html" directory in the home directories of users with the name of a specific website/web application. These users are set to the "usr/sbin/nologin" shell, which means they cannot log in to the system to manage it. This solution has the advantages mentioned in subsection 5.

2) Interpreter PHP

A service, or rather an extension to the Apache2 service, that allows the use of the PHP language by a programmer in version 8.2.4. Without this module, files with the extension ".php" would not work. Among other things, it allows for retrieving data from a database (using SQL), which is why it was installed.

3) MySQL Server

The MySQL server service allows you to create and manage databases using SQL, and is installed in version 8.3.0. Databases are managed in the terminal using SQL queries or from the outside, for example by contacting the database using PHP (subsection 2) and SQL.

4) phpMyAdmin

A service that adds a graphical menu in the form of a web application for managing MySQL databases (subsection 3). This service automatically configures itself to the website and application server we choose, in this case Apache2 (subsection 1), and also requires a PHP interpreter (subsection 2). This service is installed in version 5.2.1.

5) vsftpd

A service supporting FTP and sFTP in version 3.0.5. It allows remote management of files located on the server's physical disk. It also allows many configuration options. I used a system supporting sFTP, based on a self-signed certificate. To manage files, you must authenticate as a specific user who is on the list with the names of users who have been entered on this list (e.g. users from sub-item 1). A logged-in user is limited to files in their home directory, including the "public_html" directory, to which they can send files of their website/web application, which will be automatically available.

4. MSSQL Server

On the next Ubuntu system is SQL Server. You can connect to it using SQL Server Management Studio. It also receives requests from applications, acts as an external server for local testing. It helps in creating applications on multiple devices, as you do not have to import the database on each one.

5. PostgreSQL

Also on Ubuntu. It is accessible from the local network for every user, while from outside it only for selected users. It handles requests from applications authenticated with the appropriate user. It also allows management from the pgAdmin application.

6. Dostęp zewnętrzny

The server does not have an external IP, access from outside the network is carried out by the TailScale software. It creates a private network with tunneling, thanks to which each device has access to the others not from every location. TailScale also offers a funnel option, which allows you to launch a global domain address for a given local device. Thanks to this function, you can share websites and web applications, because it offers the possibility of sharing only port 443, 8443, 1000. I shared this function with phpMyAdmin, for friends to test.